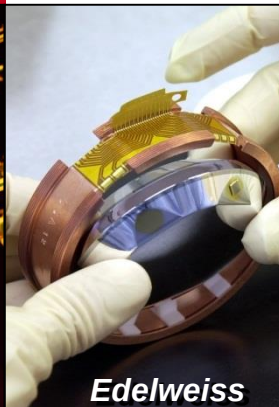


DE LA RECHERCHE À L'INDUSTRIE



# Data visualization with DATA BROWSER software

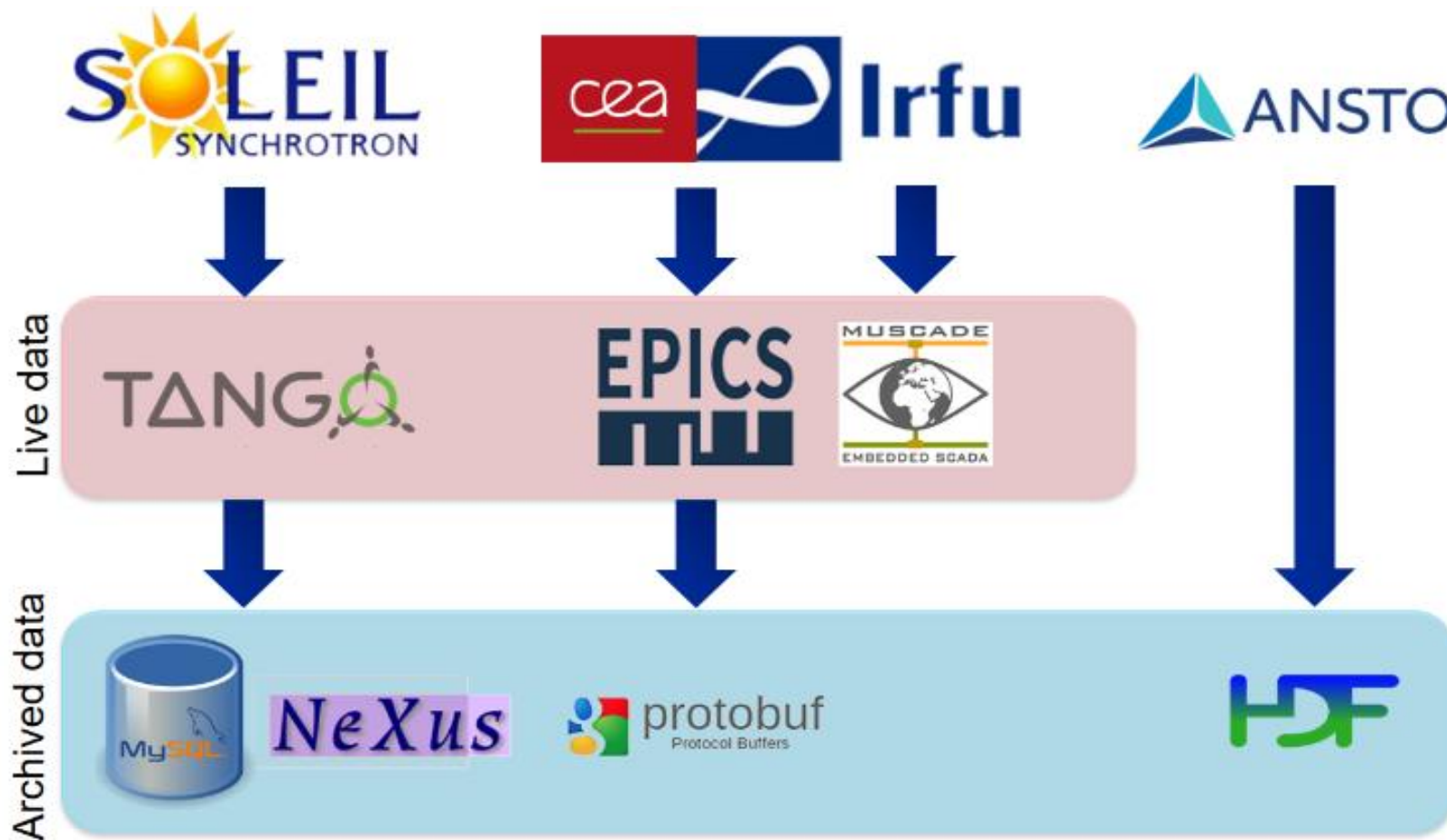


Katy SAINTIN

November 2021

Contact : [katy.saintin@cea.fr](mailto:katy.saintin@cea.fr)







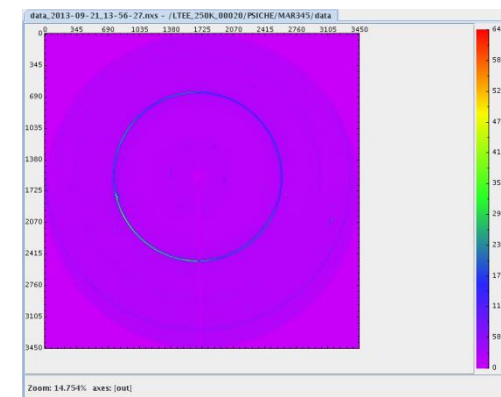
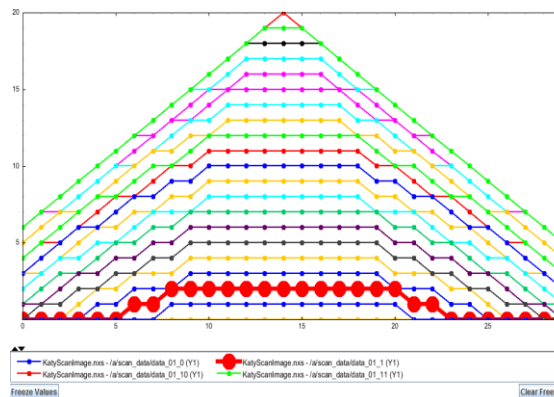
## COMETE GUI Framework

<http://sourceforge.net/projects/comete/>



Experimental frame configuration interface:

- Experimental frame ID:
- Experimental frame type:
- Tangost1/ampli:  +000.000
- Tangost1/double\_scalar:  +0000.00
- Tangost1/boolean\_scalar: ☐ false



## CDMA Framework

<https://sourceforge.net/projects/commondatamodel>



### Soleil plugins :

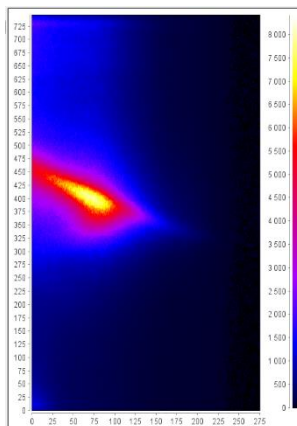
Nexus file, EDF file,  
Tango control system,  
Archiving database.

### Ansto plugins :

NetCDF,  
HDF



## COMmunity of Extensible Toolkit for Experiment



ImageViewer SWT

### COMETE Widgets

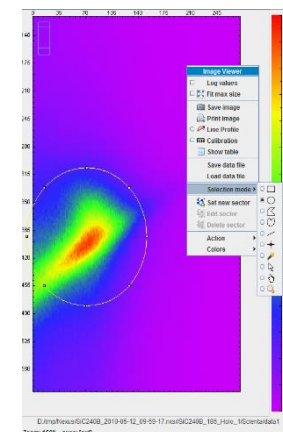
Comete  
SWT

Comete  
AWT

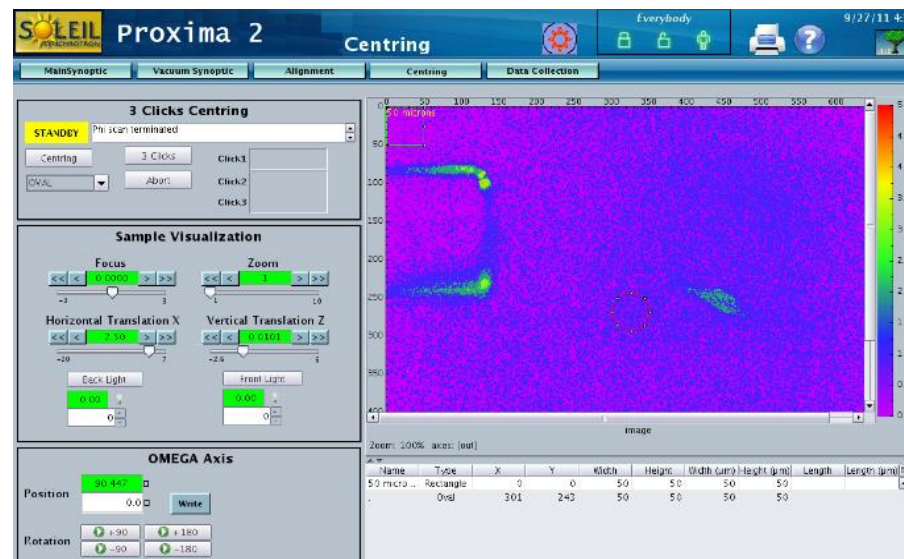
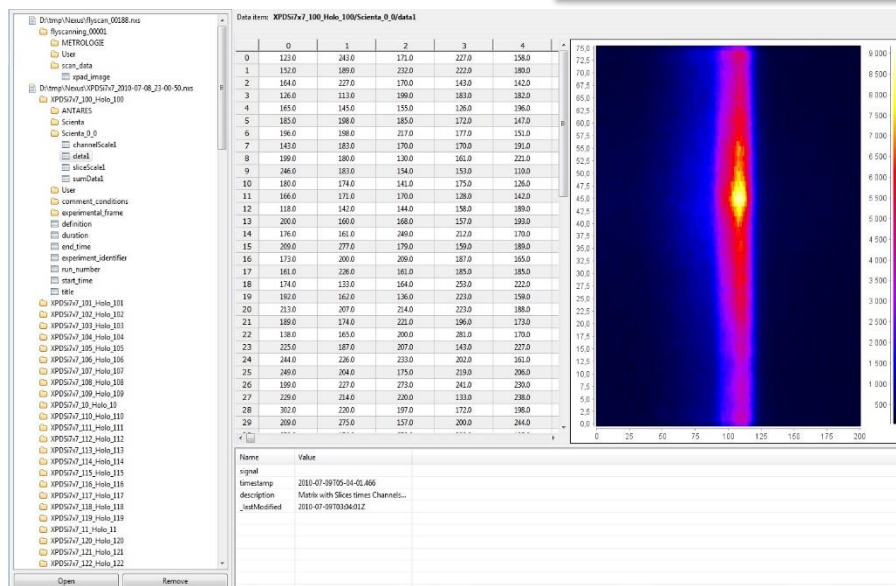
Comete  
SWING

Data Source

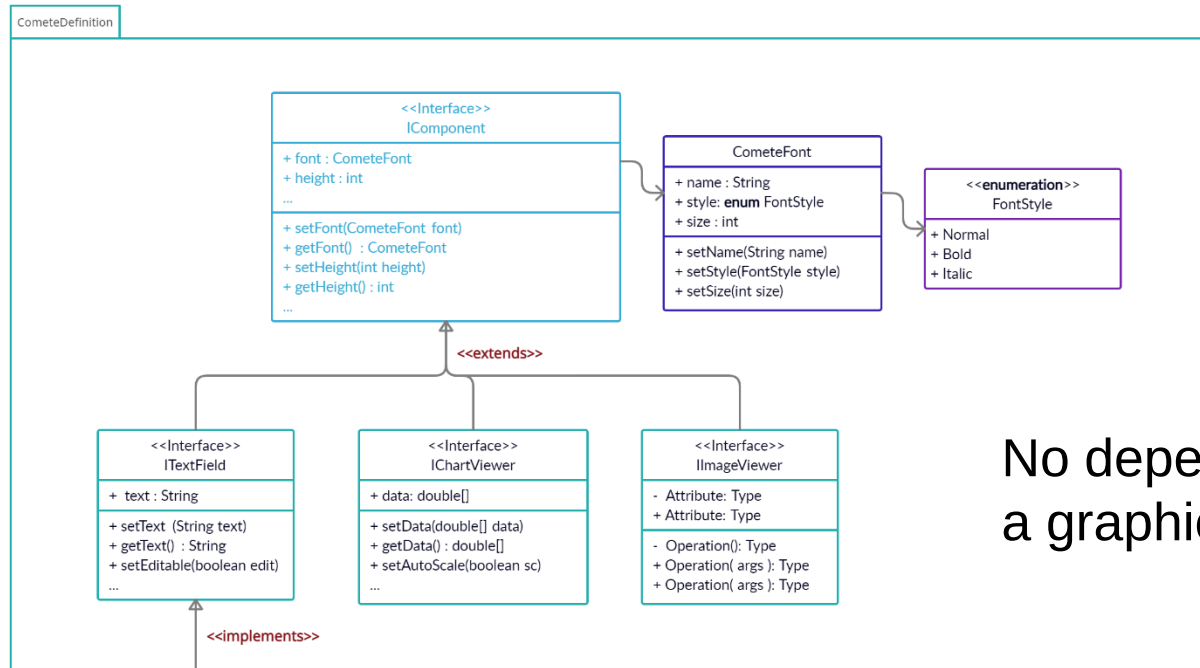
CDMA



ImageViewer SWING



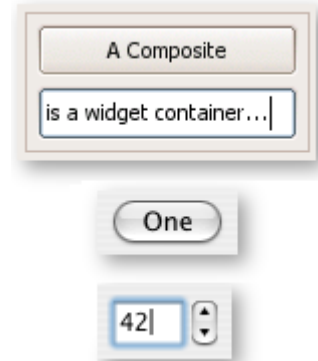




No dependency with  
a graphical framework



Graphical framework  
implementation



Common Data Model Access



Plugin



Plugin

NeXus

Plugin

TANGO

Plugin

Mambo

Plugin

EPICS

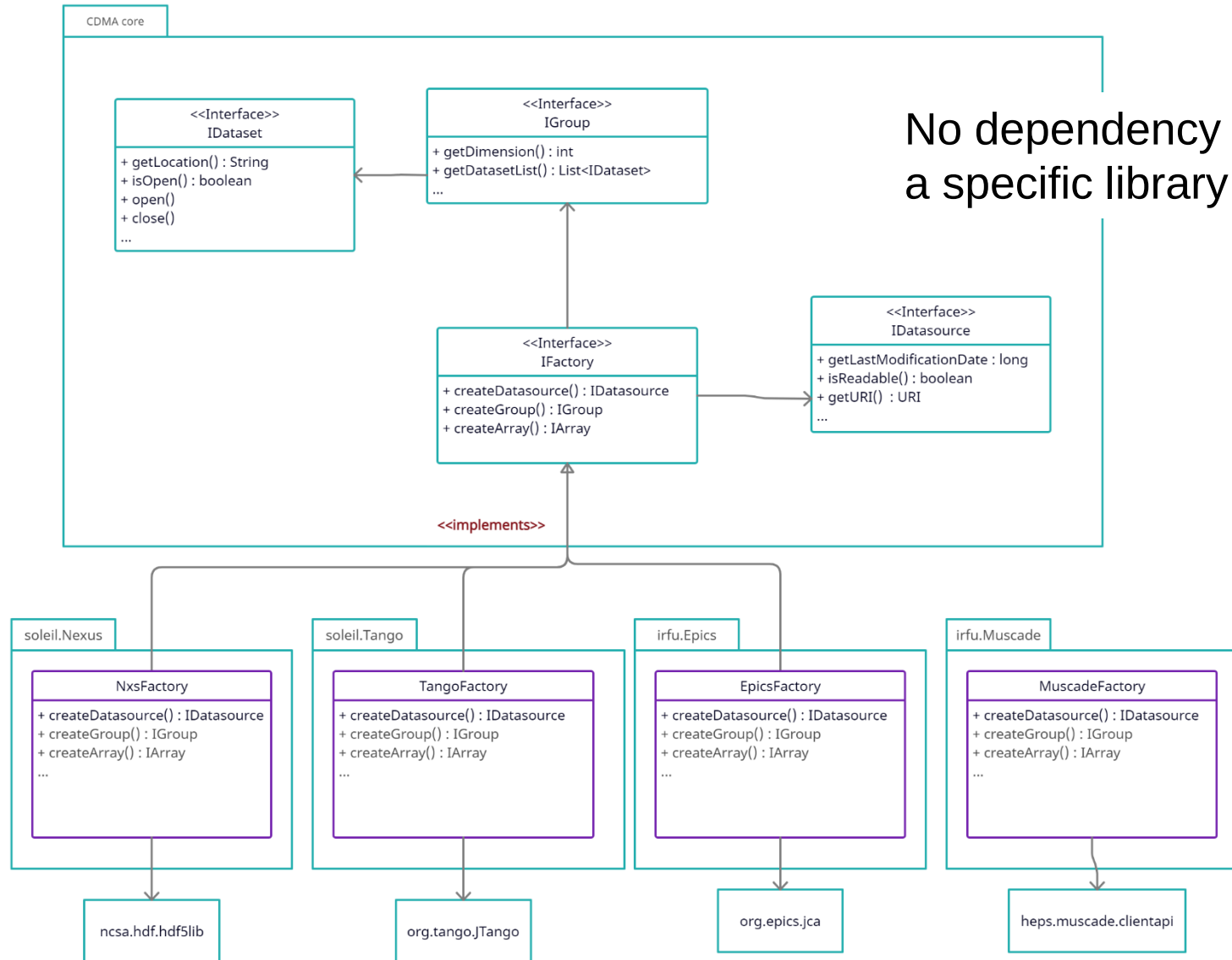


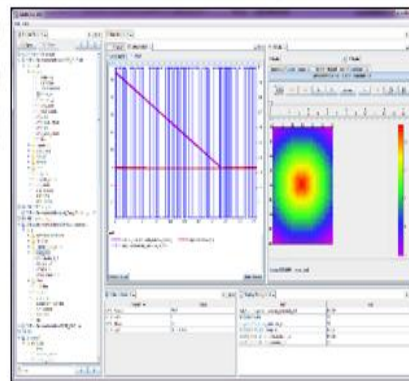
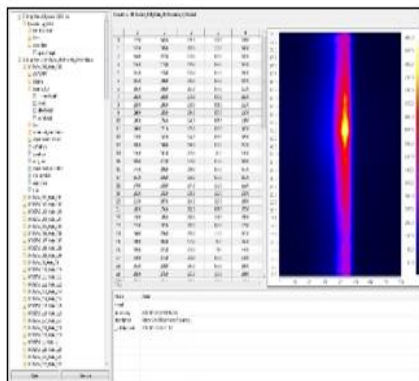
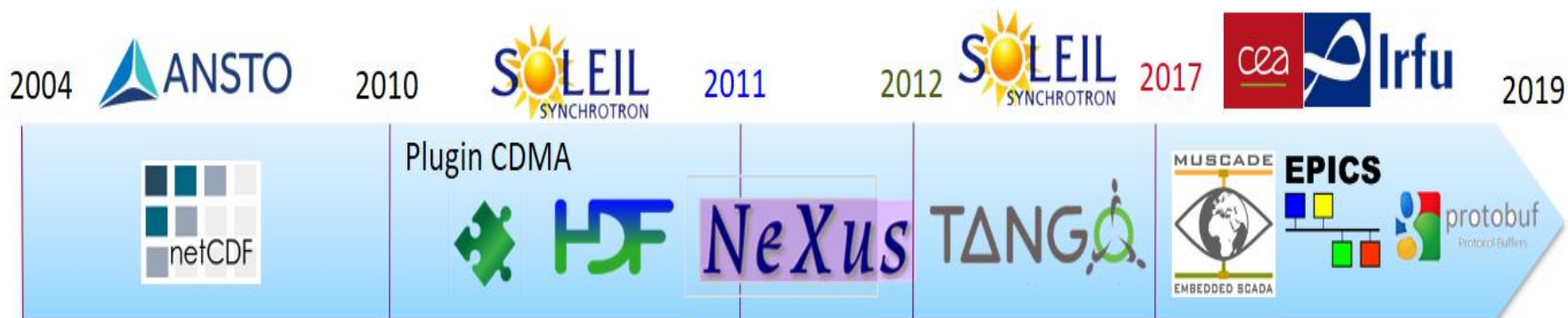
Plugin



Plugin



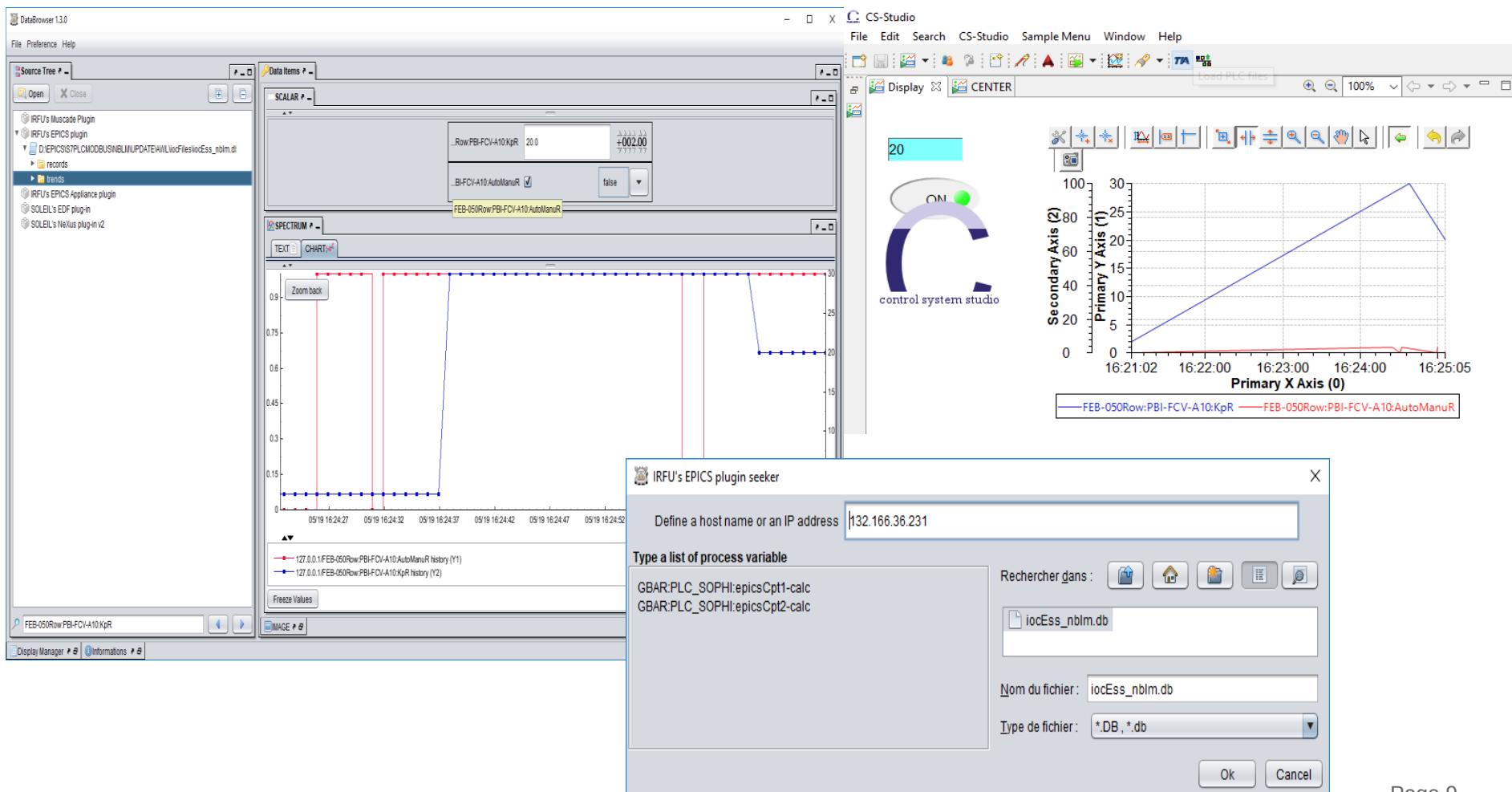




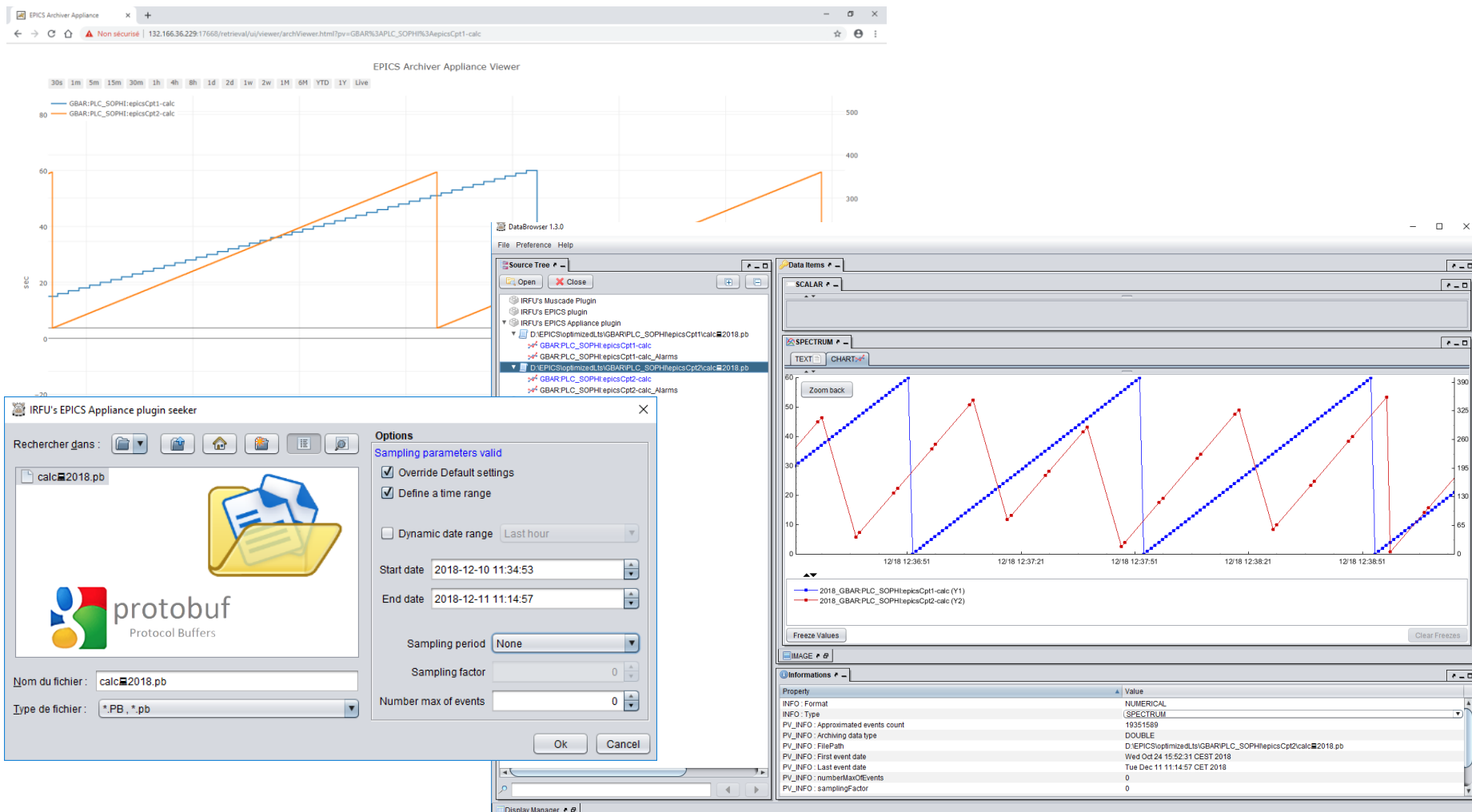


**CAJ** : Channel Access for Java to read & write PV

**VisualDCT** : EPICS Visual Database Configuration Tool for parsing a EPICS db file

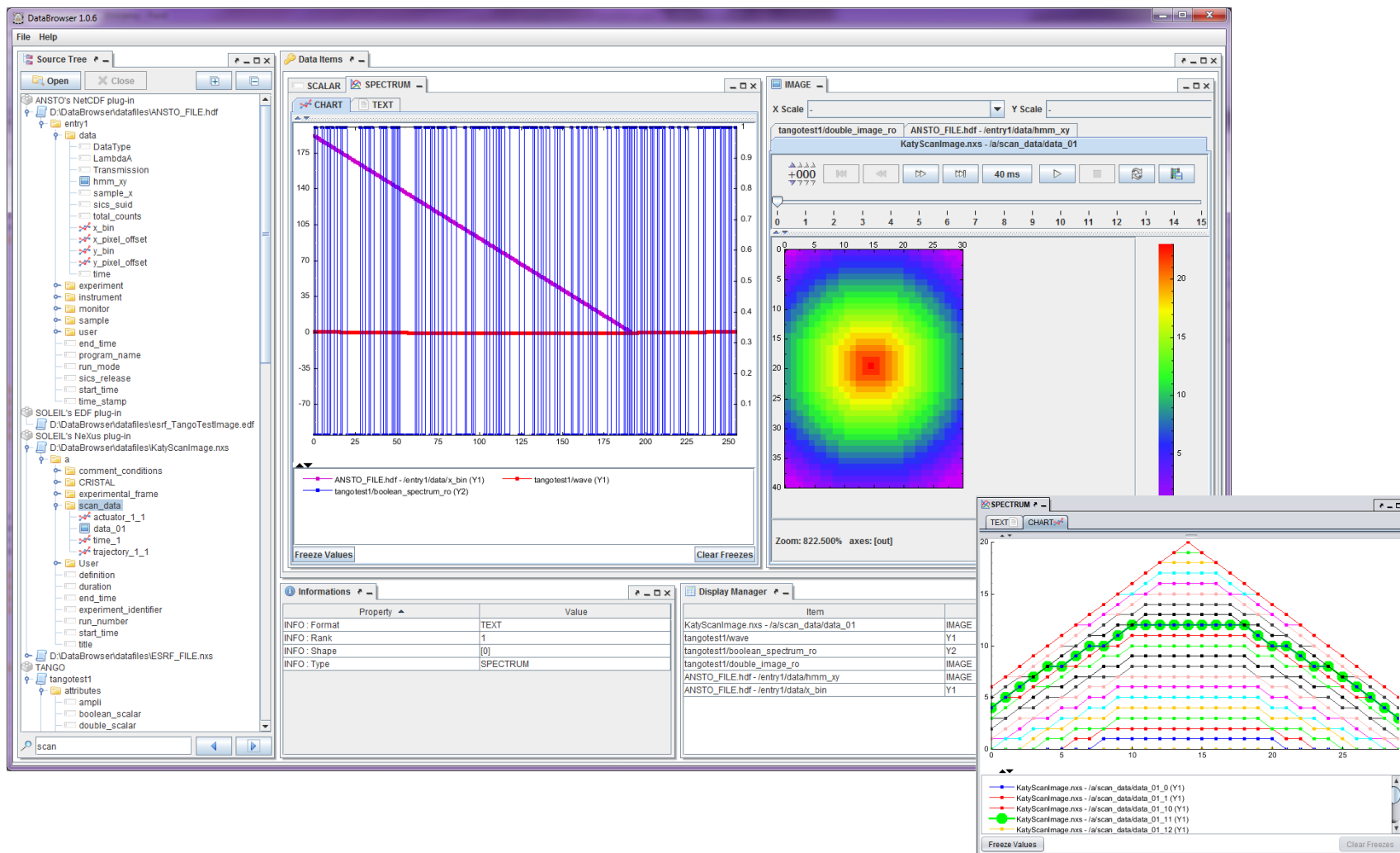


# EPICS Archiver Appliance : Read Google Protobuf files





<http://wiki.synchrotron-soleil.fr/tikiwiki/tiki-index.php?page=How-browse-downloaded-Nexus-files>  
<http://katy.saintin.free.fr/soleil/databrowser/DataBrowserCEA-1.3.1-2021-07.zip>



## Quality software – IN2P3 Formation

- Unitary test and continuous integration => Already done at SOLEIL
- Documentation programmer
- Javadoc
- Automatic deployment

## Future plugins

- MIA (Numerical Magnet Survey)
- Archiving Appliance for EPICS (Http request)
- Muscade Archiving
- HDF exportation

## REX

- SOLEIL => Nexus Format on Beam Line in off line mode
- IRFU => HDF5 (NBLM ESS), ASNET (Magnet Survey),  
futures projects as IPHI and its experiments
- New plugin => 2 weeks of development (Java Api)

